



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

**BAPS: BARANGAY APPOINTMENT SYSTEM
FOR LOCAL BARANGAYS IN
CALOOCAN CITY**

A Capstone Project
Presented to the Faculty of the Institute of Technology
Polytechnic University of the Philippines
Sta. Mesa, Manila

In Partial Fulfillment of the Requirements for the Degree
Diploma in Information Communications Technology

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ABSTRACT

Title : BAPS: Barangay Appointment System

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The distribution of appointments for various transactions is one of the most popular services that barangays offer. However, the manual appointment system used today could be more effective and more efficient. The creation of the Barangay Appointment System (BAPS) will streamline the appointment procedure and make it more convenient for both citizens and barangay officials. The primary goal of this research is to develop a "Barangay Appointment System," Residents can schedule appointments online through the BAPS, a web-based system. The appointment system will also give the barangay citizens reminders about meetings and real-time information on appointment availability and status. With an essential technical ability, all citizens can utilize and access the barangay appointment system. The researchers used the Waterfall method for software development to guarantee a systematic and cooperative approach to the system's



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development. User testing, debugging, maintenance, and professional review were used to assess the website's functional efficiency and

The data analysis revealed that the system's design and features are easy to use and efficient in helping the public request and make appointments for documents. Users of the system gave it favorable reviews.

The web-based application "Barangay Appointment System" has the potential to be used on a much larger scale by government institutions with high demands of electronic service requests.

Keywords: Polytechnic University of the Philippines, Diploma in Information Communication Technology, Web-based Application, Electronic Service, Appointment system, Waterfall methodology, system.



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Chapter 1

THE PROBLEM AND ITS SETTING

Introduction

The barangay is the smallest political unit in the Philippines, and it plays a crucial role in fulfilling and assisting the requests for various programs offered by the national or local government, such as the Pantawid Pamilyang Pilipino Program (4Ps) and scholarship grants, financial aids, and others. The barangay is also responsible for issuing and releasing critical official documents like clearances and certificates requested by their constituents.

However, applying for these programs and documents can be challenging and confusing, especially for those unfamiliar with the procedure and living far away from the barangay hall. There is also the issue of having to wait in line and then being rejected because one lacks the necessary requirements needed. This can result in an inconvenient and frustrating experience for walk-in applicants.

Recently, the introduction of QR codes has made exchanging information much more accessible with just a scan of an image. QR codes have been utilized in various ways, such as connecting consumers to a brand's website or discounts, making e-payments, joining Wi-Fi networks, and providing more information about museum exhibits.

In the case of restaurants, QR codes are being used to make appointments and reservations. Customers can easily create a reservation by scanning the code through their smartphone and filling out the generated form. The customer can also view the menu and its prices when the code is scanned, making the reservation process convenient and hassle-free.



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By using QR codes in the barangay, the process of making an appointment for official document requests can also be made more accessible. The requester can scan the QR code generated by the barangay through their smartphone, which will take them to a form that needs to be filled out with necessary information, such as the name, email address, contact number, address, and documents to be requested. After the form is filled out and submitted, the barangay will verify the request and send out the appointment schedule and the requirements, fees, and guidelines for the applied documents through the requestor's email and contact number.

Theoretical Framework

Utilizing QR codes can improve the accessibility and efficiency of public services provided by barangays. According to Carpio C. O. (2020), technology adoption can address issues like long wait times, incomplete requirements leading to rejections, and difficulties accessing information about official document processes. The assumption is that constituents equipped with smartphones are willing to embrace technology for appointment bookings and document requests.

The study examines variables such as the type of government program or document sought (e.g., Pantawid Pamilyang Pilipino Program, scholarships, financial aid, official documents) and the usage of an appointment system. Factors such as convenience and ease of application, wait times, rejection rates, knowledge about requirements and processes, distance from the barangay hall, and appointment scheduling availability are considered dependent variables. The Technology Adoption Theory explains the relationship between these variables (Straub, 2009, pp. 630-631), which delves into how individuals and organizations adopt new technologies and the influencing factors. In the context of this research, implementing the Appointment System facilitates the exchange of information between the barangay and its



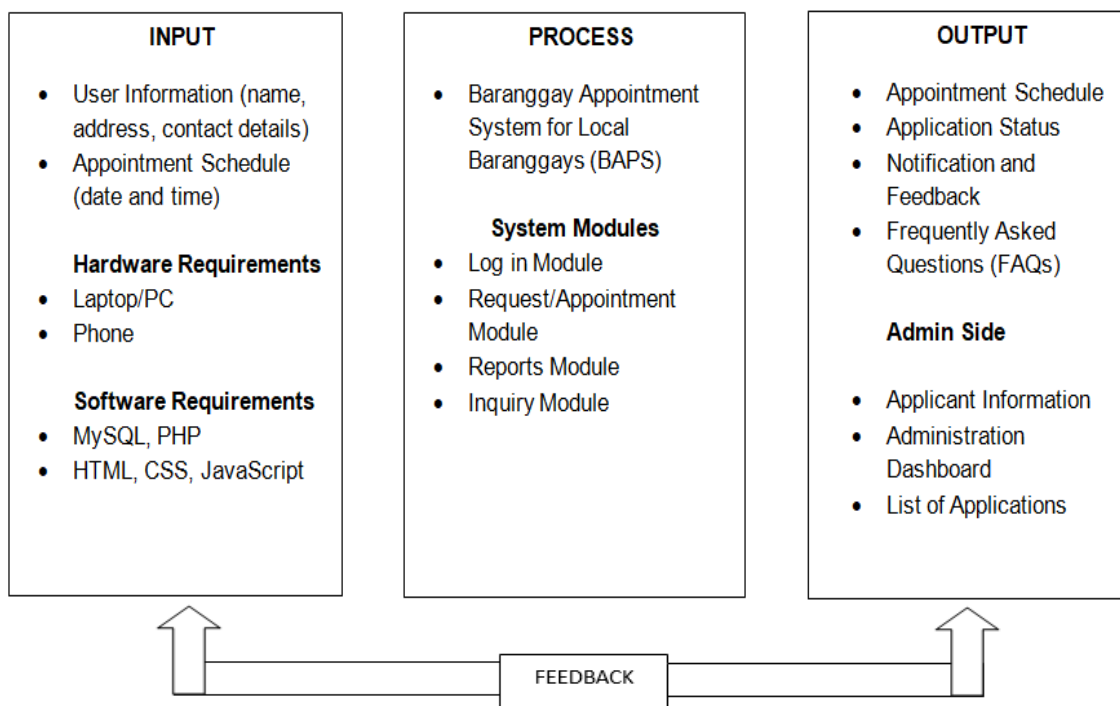
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constituents. Key factors influencing technology adoption include appointment convenience, queue elimination, and streamlining the walk-in application process. It is recognized that readiness and willingness to adopt new technologies can vary among individuals and organizations, impacting the success of adoption. Similarly, the barangay and its constituents may exhibit differing readiness and willingness to adopt the System. Successful technology adoption relies on effective communication and education regarding its benefits and user-friendliness.

This framework provides insights into the relationships among the barangay, constituents, and the usage of an appointment system in the application process. Challenges such as distance from the barangay hall and limited knowledge about document request processes and requirements can hinder constituents, resulting in rejections and wasted time. Implementing the Barangay Appointment System aims to address these challenges by providing associates with the necessary information and a more efficient request method.

Conceptual Framework

Figure 1. Conceptual Framework





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Statement of the Problem

In this study, the researchers want to answer the following questions:

1. What is the profile of the respondents in terms of:
 - a. Age;
 - b. Gender;
 - c. Type of client; and
 - d. Type of document they usually request
2. How can the Barangay Appointment System manage the queuing for clients requesting documents?
3. How does the system ensure the adaptability of the system between mobile devices and personal computers?
4. What is the level of acceptance of the respondents on the BAPS: Barangay Appointment System for Local Barangays in terms of:
 - a. Functionality
 - b. Efficiency
 - c. Reliability
 - d. Usability
 - e. Portability; and
 - f. Security

Hypothesis

Ho: By creating a profile registration index on the system, the BAPS can precisely collect all the necessary data input of the user and store it in the database.

Ha1: The admin or barangay staff will be able to properly assess the queue of user's requested documents by implementing an admin dashboard in the appointment system with the use of a database.



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Ha2: Through the development of a responsive web design that can automatically adjust depending on the user's device, the Barangay appointment system can ensure portability between a desktop and mobile phone.

Ha3: The system will collect user data feedback via a survey questionnaire that is integrated throughout the system. Once the necessary data has been gathered, the developers will assess the level of user acceptance in the overall aspect of the barangay appointment system.

Scope and Limitation of the Study

This research focuses on creating a web-based system that incorporates E-services into the barangay department's appointment system, the research also serves as the central database for the constituents. This research will focus on the enhancement and convenience of obtaining requested documents via appointment. This study, however, will be limited to users with basic technical skills and devices capable of accessing web-based and mobile browsers. Because this system is still in development, online payments and mobile applications will not be available, and the system's demonstration will be limited to the barangay within the Caloocan City, Philippines.

Significance of the Study

This study would be beneficial to the following:

To the Barangay Officials / Barangay Administrators - They will be able to manage and control all the citizens' requests properly.

To the Clients - They can easily ask for the required papers such as Barangay Clearance, IDs, and First time Job Seeker Certificate needed for their job application.

To the Future Researchers - This study will assist student researchers in being aware of and informed about the procedures involved in the numerous applications of



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QR codes. They may use it as a future reference for more studies and it would help them become better analysts.

Definition of Terms

Term	Definition
User	an individual who uses the system and has been registered on the appointment system.
Inquiry	Allows users or stakeholders to retrieve or request information from the system. It enables users to make inquiries or queries and obtain relevant data or results based on their question about the process of appointment.
User Information	Information such as educational background, employment history, personal info, and other more than has been input and stored in the database
Database	Database is where the data such as user information, users, documents and other data is stored and organized.
Appointment Schedule	An assigned time and date where a user can get the requested document in the barangay.
Log-in Module	A feature where users can input their credentials and gain access to the system
Applicant Information	Collection of user data such as name, address and contact details that will be used and verified for the requested document.



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Application Status

Refers to the phase or stage of the submitted application. The application's common statuses include "pending", "released", and "rejected".

Reports Module

A feature that provides visibility into the count of appointers and allows users to view the documents that are most frequently obtained.

Frequently asked Questions (FAQs)

Refers to the collection of the most commonly asked questions in requesting a document and their answers.

Administration Dashboard

An interface that allows system administrators to monitor and manage various aspects of the system from a centralized location.

Reports

Refers to a structured document or presentation that provides organized and summarized information of the number of users and document requester.



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Chapter 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter offers a thorough analysis of the research and literature on the subject of this capstone project. This chapter's goal is to give a comprehensive overview of the project-related research that has already been done. The review of the related literature and studies will be structured in a way that ensures a clear understanding of the key concepts and research findings that are relevant to this project.

Local Literature

Barangay Information System

Barangays are the smallest administrative units in the Philippines, tasked with providing basic services and governance to their constituents. To enhance its delivery of services, the barangay has been exploring various technological advancements that can help them streamline their processes and improve efficiency. One of these technologies, the "e-barangay," a web-based information system that seeks to digitize the Barangay's operations, was deployed in a research study titled "Towards digitalization through e-Barangay: A Web-based Barangay Information System" by Bondoc, B (2019). The barangay appointment system in the Philippines is a manual process that requires constituents to physically go to the barangay hall to apply for programs or request official documents. This process can be tedious, time-consuming, and inconvenient, especially for constituents who live far from the barangay hall. In addition, there is a high risk of errors and delays due to the manual handling of information and documents. The Philippine government has recognized the need to improve the barangay appointment system and has taken steps to modernize it. In the study by Bondoc, B (2019), the e-Barangay system was designed to provide services such as online application for



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barangay clearances, registration of businesses, and tracking of barangay finances. His system concludes that it was regarded as acceptable since respondents liked the system's features and thought it was straightforward to use.

According to “oneBarangay: An e-Government System for online transaction processing of Malanday Main Office” by Garon, J et al (2020), creating an information system helps the barangay to speed up and automate the processing of documents in order to deliver the needs of its constituents. Appointment systems have been increasingly adopted in various sectors to streamline processes, improve efficiency, and enhance customer experience. In the context of local governance, appointment systems have been used to improve service delivery and promote citizen engagement. A research conducted by Garon, J et al (2020) stated that the concept of automating and digitizing government operations has been adopted by many local governments around the world, hence the term "e-Government," and it is crucial for barangays as well. This improves operations because the data gathered through this is useful for gaining deeper insights into its constituents and visualizing such data to further improve development projects.

According to Carpio C (2020), the research report titled "Barangay Management System" , information technology application is critical for humans because computers are essential for helping people organize, store, and retrieve enormous amounts of data. It also functions as a tool that gives its users the ability to complete more activities, making it an essential item, particularly in working contexts. His research emphasizes how the government must transform to keep up with the rapid changes and complexity of the global environment. In order to encourage more active and participative political discourse and decision-making, it also emphasizes the use of information and



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communication technology (ICT) in improving government procedures and processes as well as the connections between the government, citizens, and other groups in governance. In addition, the ideal way to define e-government is as the use of information technology to manage public affairs, provide public services, and accomplish democratic governance more effectively.

However, there are challenges faced in implementing an E-government. The digital revolution could begin at the local government level, according to an article by Angara, S. In the Philippines, creating an e-government will be extremely difficult by (2021). For instance, researchers from the Philippine Institute for Development Studies noted in a September 2020 study that a lack of standardization, poor infrastructure, limited access to computers, and other related issues are impeding the creation and implementation of e-government services across the entire nation. Furthermore, He remarked that in order for us to catch up in terms of e-government services, our LGUs need to have someone in charge of overseeing the digital transformation of government operations in order to fully realize digital services. The Local Information and Communications Technology Officer Act (SB 1943) was introduced as a result. It mandates the employment of an Information and Communications Technology Officer (ICTO) in all municipalities, cities, and provinces. The following will be managed by ICTOs: developing, maintaining, and overseeing all other information and communications technology programs and services of the LGU. ICTOs will formulate and carry out digitization plans for procedures and public documents in their individual LGUs.

According to the study titled “Barangay Integrated Management System with Mobile Support” by Lim, J (2022): the benefits of using the developed system are for



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both the Barangay office and the residents. The system acts as a trustworthy instrument for the quick and automatic release of required certificates and papers, master data profiling, appointment scheduling, and more secure and precise supply management. The system offers the Barangay Health Center the quickest and most efficient way to process medical services, track medical histories, control medical supplies, and profile patient records. Additionally, the system is advantageous for the residents because it reduces the need for face-to-face interactions by allowing them to schedule appointments, access public information, and examine medical records using a mobile application. It contributes to the field of public service by presenting an effective and efficient tool to improve the delivery of services in the Barangay level as it demonstrates the positive impact of using the developed system on public service.

Web-Based Appointment System

One may frequently confront many appointments in life and business, such as selecting a convenient time to meet, whether for a job interview, business meeting, or simply to catch up with old friends. Making appointments guarantees that time will be used effectively and not wasted. Appointment scheduling "removes friction by reducing waiting time," according to Qmatic in an article titled The Guide to Appointment Scheduling. Many offices receive a lot of visitors every day; if all of these visits were walk-ins, a disorganized workplace would be unavoidable (Ampuan and Delena, 2022).

According to the study titled "An Implementation and Evaluation of Web-Based Appointment System for the Mindanao State University" by Ampuan and Delena (2022), an office can enhance the working environment by allowing clients to make appointment requests. Making an appointment allows for better staff planning and almost always



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prevents client discontent. Customers can plan their arrival close to their appointment time, which can significantly reduce their wait time and so help to avoid waiting room congestion. In this manner, a business can maintain control over the flow of clients, make the most use of its resources, and lessen or even do away with unexpected crowds. The answer that makes it simple for service providers to handle appointments is an appointment system. According to Graham Gidley's explanation in the article Patient flow management for seasonal flu and COVID-19 vaccinations, an appointment scheduling system allowed you to maintain a steady patient flow throughout the day by making slots that matched your capacity to give service available (Ampuan and Delena, 2022).

Foreign Literature

Quick Response (QR) code Technology

According to the research titled “Digitalizing Finland: Governance of Government ICT Projects” by Lappi, T et al (2019), the concept of digitalization and its impact on public sector reforms is a topic of increasing interest in academic literature. This dissertation focuses on the digitalization of the public sector, specifically the e-government transformation, which aims to improve citizen service processes through the adoption of digital technologies. His research acknowledges that ICT initiatives in the public sector are complicated and confront issues like technology uncertainty, which hinders e-government initiatives that are already dealing with strategic and regulatory constraints. A temporary project's relationship to and contribution to the strategic goals of a more permanent structure are the main topics of project governance.



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According to a research paper titled "Smart Appointments Booking Systems for Social Distancing: Technologies and Solutions" by F.H. Almuhsin et al. (2021), studies on the use of QR codes in appointment booking systems have shown that many such systems have been developed to address the issues inherent in traditional systems, which require clients to be physically present and to wait for their turn based on a numbered paper ticket. However, the spread of COVID-19 has rendered these systems useless. No solutions for the appointment scheduling systems in the current situation with requiring social seclusion to lessen the transmission of COVID-19 were discovered through a review of studies. Additionally, in the present COVID-19-related environment, QR code apps and social distancing strategies offer answers for traditional appointment scheduling systems to function more effectively. In this new normal where people should wear masks all the time and follow safety protocol health guidelines, an QR-code based appointment system in the local government unit will be an advantage in fighting and preventing the spread of COVID-19, and other infectious diseases. With this type of system implemented, paper processing will be much more effective with the benefit of citizens being safe from health hazards resulting in an increase satisfactory in overall safety of the people. Technology has improved thru the years, billions of people in the world own a smartphone with the latest model and features inside of it, because of that features such as the Quick Response or QR code for short has been more utilized much efficiently. Now almost everything is one click away through using QR code. A person can order food, buy appliances, browse for clothes and many more things are now much easier and citizens are still yet to have a better service in the government field.

A research study titled "Integration of Web-Based and Mobile Application with QR Code Implementation for the library Management System" by Marina, et al (2021)



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the QR code technology is low-cost, convenient, easy to implement, and easy to use, there is no doubt that in the future it will be expanded to other areas of interest. QR code has been used in lots of different ways but has one goal, to provide a much convenient and efficient way of service, and with being said other interest such as the use of QR code technology to the government field will be beneficial, not only it will be cheap and low-cost but it will also boost the efficiency of service of our local barangay agency which will lead to better satisfaction of citizens and improved overall trust in our local government unit.

Online Students' Appointment System

According to study titled "Online Students' Appointment System for University Administration" by Abu, Z et al (2022, it improved the university administration's efficiency by decreasing the waiting time for the students. Additionally, the system provided students an easier and more comfortable way to communicate with university administrators. According to the research's findings, institutions can provide high-quality services to students with the use of an online appointment system. This research study is relevant to the current research study since it provides insights on the features of an appointment system that can provide convenience to the constituents. This study relates to the research by Zamri, K. N. N. B. K., & Ishak, W. H. W. (2022).

According to the research titled "Design and Development of Web-Based Appointment System" by Zamri, K. N. N. B. K., & Ishak, W. H. W. (2022) mentioned that in order to improve scheduling and reduce human error that results from manually creating appointments, appointment systems can be employed. Users can also participate actively in the appointment-making process by selecting the time and day that



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work best for them. An appointment system is necessary, especially during pandemics when maintaining strict safety standards and social distance is required due to the serious threat of viral spread.

According to Abu, Z. et al. (2022), the 50th percentile average System Usability Scale (SUS) score is 68. Anything with a SUS score of 68 or more would be deemed above average, and anything with a score of 68 or lower would be deemed below average. The research study by Abu, Z et al (2022) showed that the Online Students' Appointment System at Administration University achieved an overall mean SUS score of 68.9, indicating an 'above average' degree of usability acceptance.

Online/Web-Based Appointment System

A study titled “Effect of an online appointment scheduling system on evaluation metrics of outpatient scheduling system” by Habibi, M. R. M. et al (2022) applying web-based appointment systems solves issues concerning the traditional system. Manual appointment-making procedures frequently resulted in overbooking or failure to cancel an appointment in a timely manner, leaving space for another to be scheduled in its place. Longer wait times for clients may result from the problem. An appointment system can also be used to improve scheduling to reduce human error that results from manually creating appointments. Customers can participate directly in the appointment-making process by selecting the time and date that work best for them. A queue management system may be integrated to the system, allowing businesses to seamlessly transition customers from online to in-person encounters. It can enhance the standard of customer service by reducing wait times.



P O L Y T E C H N I C U N I V E R S I T Y O F T H E P H I L I P P I N E S

To automate and organize their appointment calendar, professionals, enterprise merchants, banks, and other businesses utilize an online appointment scheduling system, also known as an electronic appointment scheduling system or an online booking system (Osbourne, 2022). According to the research of Osbourne (2022) titled “Online Appointment Scheduling”, appointments had to be made over the phone in the past. For everyone concerned, it was a time-consuming procedure. The retail employees had to manually arrange every booking due to the overwhelming number of calls and emails for appointment requests.

When it comes to making a reservation, customers have greater freedom with electronic appointment scheduling. For example, most systems let your clients book appointments using your website, mobile app, search engine profile, QR code, social media, and more. Additionally, there is no curfew; clients can schedule an appointment at any time of day (Osbourne, 2022).

Based on the article by Osbourne (2022), the appointment scheduling procedure can be automated to save load on your store personnel. They get back many of the hours they would have spent on the phone setting up appointments, and they can now devote that time to duties that are more important. Because modern consumers like retailers and banks who employ digital technologies, you are also more likely to gain new clients than your rivals.



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Chapter III

METHODOLOGY

Research Design

During the system design phase, a comprehensive plan for the barangay system is developed based on the requirements gathered during the earlier phase. This plan includes several essential details such as a data flow diagram, software design, logical design, and physical design/specifications. The data flow diagram describes the flow of information within the system, while the software design outlines the overall structure of the software. Additionally, the logical design identifies the different components of the system, such as the user interface and database, and how they interact with each other. The physical design/specifications, on the other hand, define the hardware and software requirements necessary to implement the system. Together, these components provide a detailed blueprint for the development of the barangay system, ensuring that it meets the needs of end-users while adhering to industry standards and best practices.

Software Development Methodology

During the implementation phase, the developers will construct a program that adheres to the specifications and requirements outlined in the previous phase. Following this, the system will move to the testing phase, where it will undergo rigorous evaluation to ensure that it meets the desired level of service as defined in the earlier phase. Once testing has been successfully completed, the complete software system will be deployed in the barangay for end-users to utilize and provide feedback.

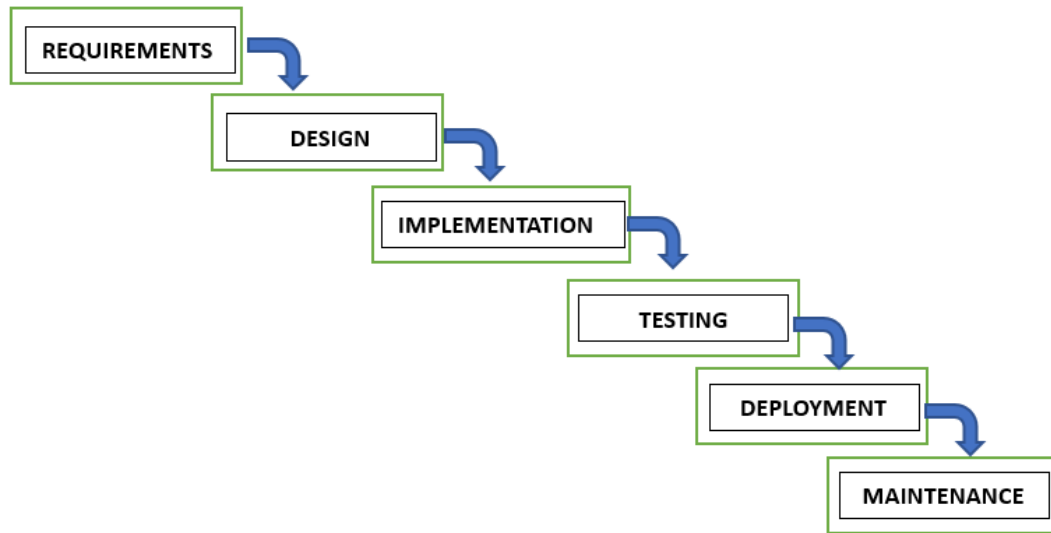


Figure 2. Waterfall Model

The maintenance phase will then commence, whereby researchers will continually monitor the system's performance, technical errors, and bugs while collecting end-user feedback. Based on this feedback, developers will then apply fixes and updates to ensure that the software continues to function appropriately until the end-users are satisfied.

Population

The target population for this study is the residents of Barangay 177, Camarin, Caloocan, including students, employees, and unemployed individuals. This specific geographic area is the focus of the study, encompassing all individuals under 18 and above in the barangay regardless of their employment status.

The selection of this target population is driven by their frequent document requests. By studying this group, researchers can gather valuable insights into their



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document-related needs, challenges, and preferences. This information can enhance document-related services and processes within the barangay.

Sample Size

The research will utilize a sample size of 100 participants to ensure a focused and concise evaluation of the system. This sample size has been determined to provide sufficient data for meaningful conclusions while considering the resources and constraints of the study. The aim is to obtain precise results for the system evaluation by allocating ample attention and detail to each participant. Considering the limitation of time if the number of participants was to be increased, an alternative sampling technique called random sampling can be advantageous. The research can achieve a fair and reliable survey outcome by random selection of participants, allowing for a representative sample without the need for substantial increases in participant numbers.

Sampling Technique

The chosen sampling technique is random sampling. This technique involves randomly selecting participants from Barangay 177, Camarin, Caloocan City who are 18 years old and above. The rationale for selecting individuals from this age group is that they are the ones who typically request documents in the barangay.

By utilizing random sampling, we aim to gather a representative sample from the population of interest. The random selection process ensures that each eligible individual in the barangay has an equal chance of being included in the survey. This approach allows us to obtain a diverse range of perspectives and experiences, which can contribute to a more comprehensive evaluation of the project. The feedback and



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responses provided by these participants will be valuable in enhancing the system. By incorporating the voices of the individuals interacting with the system firsthand, we can gain valuable insights that will ultimately contribute to its effectiveness and usability.

Sources of Data

The researchers gathered information through conducting interviews with the barangay officials in Local Government Units. The researchers ask in-depth questions about barangay forms, documents, and proper services. The researchers will also provide a letter of consent to the participants for they be also to be aware of the survey that was mentioned about.

Research Instrument

An interview with a guided questionnaire will be used in collecting valuable information that will add to the researchers on developing the proposed system. The questionnaire contains 10 questions about barangay documents, forms and other common barangay services that are needed in day-to-day work in the office. It's a type of questionnaire where participants need to answer factually and accurately.

Data Generation Procedure

The aim of this research is to aid the barangay sectors in providing services for their residents. With the help of the Barangay Appointment Systems (BAPS) requesting documents from the barangay would make it much more convenient for the residents of the local community. The data collection method that the researchers used to gather data is by handing out questionnaires/survey containing the important inquiries needed to gather data from the local barangays.



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To ensure that everyone in the research team gathers data in a uniform manner, such as by conducting experiments under the same conditions and using objective criteria to record and classify observations, standardizing methods entails outlining specific, step-by-step instructions. This helps prevent common study biases including information bias and prejudice resulting from omitted variables.

By using this method, the accuracy of the collected data can be assured, and can also be utilized for future research. To collect data about the needed information to apply for a certain document from the local barangays, the researchers conducted a survey with open-ended questions to a sample of 5 local barangays and a representative to answer the questionnaire across different locations.

Ethical Considerations

In this research, data collection is conducted using two widely employed methods: face-to-face interviews and online surveys. Both methods are utilized due to the research's focus on determining the average daily number of citizens requesting documents in three barangays, as well as performing a survey in the barangay with the highest daily requests.

To ensure the sample of barangays selected for the research accurately represents the population, the researchers have chosen to include citizens from Barangay 177 in Camarin, Caloocan City. The selection of this barangay was based on rigorous and systematic criteria, which considered factors such as demographic profile and other relevant aspects that align with the research objectives.



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To initiate the research in each barangay, the researchers have sought the permission of the respective barangay captain. The ethical principles of informed consent and respect for the autonomy of the participants have been observed, and the researchers have provided a formal letter of intent, signed by their professor, that outlines the purpose, methodology, and scope of the study. The researchers have also presented a valid identification document to confirm their identities and credentials, thereby establishing the credibility and legitimacy of their research endeavor. The adherence to the prescribed ethical and professional standards and procedures is paramount in ensuring the validity, reliability, and generalizability of the research findings. We apprised the interviewee that our study aims to provide e-services to the residents of the barangay, particularly in addressing their document-related concerns. The study intends to bring about greater convenience to the residents, given the current surge in population, by mitigating the need to queue in long lines on a daily basis.

Statistical Treatment of the Data

Data collection is essential for any research project. The raw data collected will be processed, organized, and analyzed to make inferences about a population, test hypotheses, and identify patterns in data to convey meaningful information. The researchers used the following statistical method to interpret and evaluate the data:

1. Frequency and Percentage Distribution

In order to correctly assess and summarize the gathered data, frequency and percentage distribution will help in displaying the statistical percentage of each data point or grouping of data points. The total number of observations within each data point or group of data points must be counted, and the total number of observations must then be divided by each data point or group of data points to produce the



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frequency and percentage distribution. It will assist researchers in expressing the relative frequency of data in the collected respondent profile.

Formula:

$$\% = \frac{f}{N} \times 100$$

where: % = percentage

f = frequency

N = total frequency

2. Mean

For symmetrical distributions, the mean is the statistical tool for determining the central tendency. By summing all of the values in a data collection and dividing by the total number of values, the mean is calculated. This metric will be used to interpret the average level of agreement on a Likert scale in order to determine how efficient the barangay appointment system's e-services are.

Formula:

$$\bar{x} = \frac{\sum x}{n}$$

where: $\bar{x}$ = mean **n** = total number of values



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$\sum x$ = sum of all values

3. Range

Range is a statistical tool for measuring variability and describing the spread of a data set. The range is derived by taking the data from lowest to greatest value and subtracting them to compare the spread of two or more data sets. This tool will be used to compare the level of agreement on the Likert scale, displaying the comparison of data in the two categories, whether it falls in the strongly agree or strongly disagree scale.

Likert Scale

In this capstone study, participants will be asked to provide their assessment of a series of statements or items on a scale indicating their level of agreement or disagreement. The Likert scale will specifically be utilized to analyze resident satisfaction in Barangay 177, Camarin, Caloocan City. The options on the scale are "strongly agree" and "strongly disagree."

Each issue in the questionnaire has prepared response alternatives, allowing respondents to pick their choice. The lowest and maximum values of the 5-point Likert scale were carefully calculated by the researchers. A difference of 4 was found by calculating the range, which is the difference between the minimum value (1) and the greatest value (5).

This difference was divided by the total number of points on the scale (5) to establish the interval value, resulting in 0.80. Multiplying this interval value by the lowest value (1) yielded the maximum value of the scale.



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To effectively gather information, the study will leverage six ISO standards: function stability, performance efficiency, portability, reliability, usability, and security. These standards will serve as a framework for assessing various aspects of the system. To ensure the questionnaire's quality and obtain satisfactory responses from the participants, Mr. Marc Anthony Bautista verified the questionnaire.

For Questionnaire

Definition	Likert Scale	Interval
Strongly Disagree	1	1.00-1.80
Disagree	2	1.81-2.60
Neutral	3	2.61-3.40
Agree	4	3.41-4.20
Strongly Agree	5	4.21-5.00

Table 1. Rating and Interpretation for Questionnaires

Data Case Analysis

Our research adopts a descriptive mixed method approach to guide the development of the Barangay Appointment System. Primary data is collected through distributed forms from the public servant and questionnaires completed by officials from three selected barangays during document processing. These primary data sources serve as vital resources for system design, providing essential information for development and implementation. Our research paper incorporates secondary data from



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both foreign and local studies, serving as a foundational element. These secondary sources offer a comprehensive understanding of the evaluated data, contributing to the enlightenment of our research findings. They provide a broader perspective, integrating insights from diverse studies conducted in various contexts.

By employing the descriptive mixed method approach and integrating primary and secondary data sources, our research aims to achieve a comprehensive analysis. This approach ensures that our research paper benefits from empirical data collected directly from residents, coupled with a broader knowledge base derived from existing studies. Consequently, it enhances the reliability and validity of our findings.

System Architecture

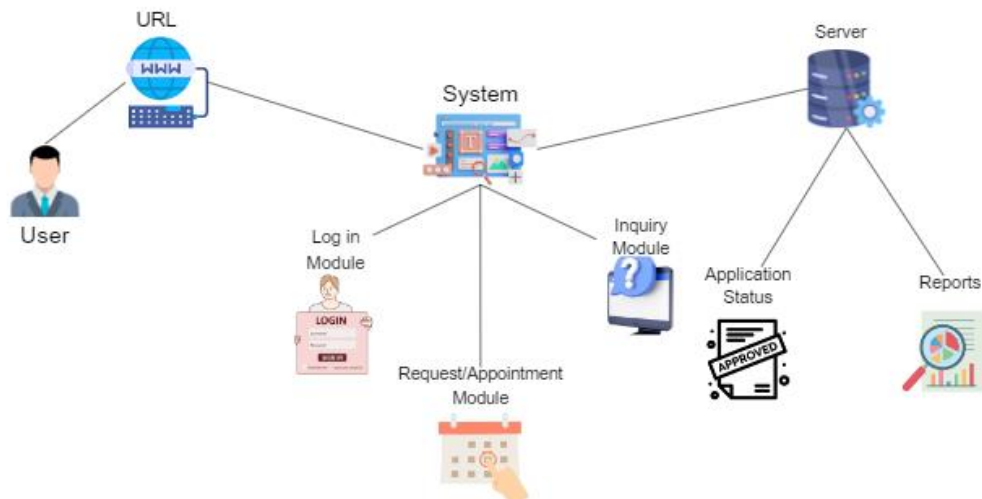


Figure 3. Proposed System Architecture



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Chapter IV

PRESENTATION, ANALYSIS AND RESULTS

This chapter encompasses a comprehensive overview of the information, incorporating the data collected, the outcomes derived from statistical analysis, and an insightful interpretation of the results based on the feedback shared by respondents in the survey. The data, which aligns precisely with the questions posed in the issue statement, is presented in various formats such as tables, graphs, and textual descriptions.

Table 2. Hardware Components

The table below shows the hardware components used in creating BAPS.

Hardware Used	Description
Desktop Computer	The desktop computer is the primarily device that has been used by the developer to create the website. The computer processes all the vital components such as the software tools and application that is responsible for the stability of the system.
Network Connection and Equipment	Router and LAN cable is the hardware equipment that will aid the up time of the system in the web. After completing the system, the researcher will use web hosting to connect the appointment system through the network and be available in the internet.
Storage Devices	The barangay appointment system stores its data physically on the hard disk drive on the desktop computer.
Server	The server provides functionalities on the system such as providing of services, sharing, storing and managing data resources. The barangay appointment system utilizes the relational database management system (RDBMS) to provide e-services to its clients.

Table 3. Software Components

Table 2 shows the software components and programming language used in creating BAPS.

Software Used	Description
Operating System	Windows 10 is the operating system that is used on developing the program for the barangay appointment system.
Visual Studio Code (HTML, CSS, PHP,	Is the IDE tool that has been used for developing, coding, debugging, and deploying the appointment system.



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JavaScript)	The barangay appointment system's structure, embedded multimedia, and format are all created using HTML (HyperText Markup Language). <u>CSS</u> (Cascading Style Sheets) is used in customizing the graphical design of the system. <u>PHP</u> is the programming language used for the appointment system's server-side web development. Its primary function is to generate dynamic web pages, as well as to manage form data and databases. The researchers used <u>JavaScript</u> as the programming language to make the web pages of the barangay appointment system interactive. This language's main functionalities are functions for form validation, interactive components, and dynamic content updates.
Xampp (MySQL, phpMyAdmin)	XAMPP (cross-platform, Apache, MySQL, PHP, and Perl) is the primarily web server application that is used to interpret the programming languages that has been developed the web-based system. MySQL served as the Barangay Appointment System's main database, and it is used to control and save data for the system. phpMyAdmin is the application that provides the appointment system with a graphical user interface for tasks such as table creation, query execution, and permission management. This web application is a tool that will be used on the system's admin side to improve the efficiency of services in the requested documents..

Software System Implementation

The researchers carefully analyzed the needs of the users during this phase in order to provide a quality software design that will help improve the user experience. The researchers considered the availability of resources, the cooperation of stakeholders, and the questionnaire responses on implementing the software system.

To increase the effectiveness and efficiency of user participation in the appointment system, this involves offering instructions, establishing intuitive functionalities, and building a user-friendly interface.



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Detailed Procedures

The following general processes should be accomplished.

Research and Data Gathering

The researchers can identify the specific challenges and needs of the barangay constituents by conducting extensive research and data collection. After taking into account user comments, researching related systems, pinpointing fundamental issues, and locating pertinent data sources, the developers can come up with fresh ideas and insights to further enhance the efficacy and efficiency of the barangay appointment system. This process also takes into account legal and regulatory requirements, as well as data privacy.

Analyzing and Planning

A comprehensive plan for the system is developed based on the identified requirements after extensive research of existing studies similar to the system and user feedback. The design of the appointment system is meticulously planned, as are its portability, reliability, security, and performance. The user interface and database are carefully designed to be aligned and to create a flow that allows the two components to interact smoothly. Testing, debugging, and implementation strategies are developed to ensure the system's quality performance. Plans for allocating resources for deployment and maintenance are established to ensure the appointment system runs smoothly. Taking all of these factors into account, the researchers created a dynamic and user-friendly barangay appointment system.

Programming and Developing

During this phase, the researchers utilize the acquired information from the conducted research to proceed. Suitable programming languages, frameworks, and tools were chosen based on the predetermined requirements of the system to develop



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the system's elements. Through the help of the interviews conducted by the researchers, the gathered data was incorporated into the planning of database structure, and then into the system. By making use of coding techniques and continuously treating the process, producing the system's user interface design. Lastly, the researchers brainstormed to create the system's process flows, user interface, and data summary reports.

Testing

During the testing phase, the Barangay Appointment System goes into a thorough evaluation and scanning where its functionality, reliability, and performance are assessed by the researchers. Test runs are conducted to validate the system's capabilities such as the filing of requested documents, notifications, and status reports of the transactions. Any problems or errors discovered during the testing are documented and resolved by the researchers. Through this rigorous testing process, we verify that the system coheres to the specified requirements and runs smoothly, producing a sturdy and reliable Barangay Appointment System that reaches the user's desired outcome.

Debugging

The researchers thoroughly examine and resolve any problems discovered in the barangay appointment system. They utilize their research expertise to carefully investigate and determine the underlying causes of bugs and abnormalities. By applying systematic debugging methods such as code inspection, inspecting elements, and error tracking, the researchers pinpoint specific lines of code or logic that needs utmost attention. Working together, the researchers implement effective solutions and adjustments to fix the identified issues. This process enables the researchers to efficiently identify and resolve problems, ensuring that the Barangay Appointment System operates smoothly and remains reliable.



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Implementation

Before implementing the system, the researchers integrate and validate the hardware and software components to ensure that the system is operating smoothly and properly. This rigorous process aims to effectively address user needs and deliver a reliable solution. It involves substantial effort, including extensive code learning and gaining an understanding of how the components operate.

Project Investigation

Profile of the Respondents

Table 4. Gender of Respondents

Gender	Frequency	Percent
Female	36	36%
Male	64	64%
Total	100	100%

Table 4 presents the gender-based demographic profile of the respondents. The male respondents constitute the majority, accounting for 64% of the total, while the female respondents make up the minority, representing 36% of the total.

Table 5. Age of Respondents

Age	Frequency	Percent
18-24	72	72%
25-34	21	21%
35-44	3	3%
45 and above	4	4%
Total	100	100%



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Based on the table, it can be observed that the most frequent age group among respondents who typically request documents in the barangay is 18-24, constituting 72% of the total. Following that, the next significant age group is 25-34, representing 21% of the respondents. The age group of 35-44 forms a smaller proportion with 3%, while individuals aged 45 and above account for 4% of the total.

Table 6. Employment Status of the Respondents

Employment Status	Frequency	Percent
Student	58	58%
Job Seeker	9	9%
Employee	28	28%
Barangay Officials	5	5%
Total	100	100%

The employment status of the respondents was categorized into three groups: students, job seekers, and employees. Out of the total sample size of 100 respondents, 58 individuals (58%) identified themselves as students, 9 individuals (9%) as job seekers, and 33 individuals (33%) as employees. Therefore, the majority of the respondents were students, accounting for 58% of the total, followed by employees at 33%, while job seekers constituted the smallest proportion at 9%.

Table 7. Documents they usually request

Document	Frequency	Percent
Barangay Certificate/Certificate	82	82%
Barangay Indigency	53	53%
Barangay Identification	50	50%
Business Clearance	26	26%

With the sample size of 100 respondents. 82% of the total respondents usually request for the Barangay Certificate/Clearance, 53% for the Barangay Indigency, 50%



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for the Barangay Identification, and lastly, 26% for the Business Clearance. Therefore, the most frequently requested was the Barangay Certificate/Clearance, accounting for 82% of the total, followed by Barangay Indigency at 53%, Barangay Identification at 50%, while the Business Clearance constituted the smallest proportion at 26%.

Table 8. Respondents' Assessment of the System in terms of Functionality

FUNCTIONALITY	Standard Deviation	Mean	Verbal Interpretation
1. The system's user interface is easy to understand and use.	0.587668	4.59	Strongly Agree
2. The BAPS's features and functionalities are all present and working.	0.632136	4.38	Strongly Agree
3. The system provided all of the information required to correctly complete the forms.	0.673525	4.47	Strongly Agree
4. The system efficiently managed the clients queuing to reduce waiting time.	0.656283	4.44	Strongly Agree
5. The system efficiently controlled the client's queue to reduce processing and waiting time.	0.573048	4.57	Strongly Agree
6. The system successfully addresses all the designated tasks and fulfills the objectives set by the users	0.662029	4.31	Strongly Agree
General Weighted Mean:		4.46	Strongly Agree

Based on the data presented in the table, we can conclude that out of the 100 respondents, the average score obtained was 4.46. This average score is associated with the verbal interpretation of "strongly agree," indicating that the majority of the respondents expressed a high level of agreement and satisfaction with the Barangay



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Appointment System. These findings suggest that the respondents generally hold

EFFICIENCY	Standard Deviation	Mean	Verbal Interpretation
1. The BAPS handled your appointment scheduling efficiently and effectively.	0.576562	4.53	Strongly Agree
2. The system provided a convenient appointment schedule for requesting and claiming documents.	0.643224	4.48	Strongly Agree
3. The BAPS can complete the required task at a sufficient level without any technological issues.	0.693112	4.38	Strongly Agree
General Weighted Mean:		4.46	Strongly Agree

positive perceptions towards the system and are highly satisfied with its functionality.

Table 9: Respondents' Assessment of the System in terms of Performance Efficiency

Analysis of the data provided in Table 9 reveals that among the 100 respondents, an average score of 4.46 was obtained. This average score, corresponding to the verbal interpretation of "strongly agree," signifies that a significant majority of the participants expressed a substantial level of agreement and satisfaction with the Barangay Appointment System. These findings strongly indicate that the respondents generally possess positive perceptions of the system and exhibit a high level of satisfaction with its efficiency and performance.

Table 10. Respondents' Assessment of the System in terms of Security

SECURITY	Standard Deviation	Mean	Verbal Interpretation
1. The security functionality of the BAPS system has a trustworthy authentication technique for user data protection.	0.686375	4.44	Strongly Agree



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2. The BAPS system has a function feature to ensure the user's information credibility.	0.719919	4.37	Strongly Agree
3. The BAPS has security measures for fraudulent activities such as forgery and identity theft.	0.739847	4.41	Strongly Agree
4. The BAPS implements strong access controls and authentication mechanisms to prevent unauthorized access.	0.673	4.46	Strongly Agree
5. The BAPS system clearly expresses its terms and agreement on using the website.	0.917341	4.37	Strongly Agree
General Weighted Mean:		4.41	Strongly Agree

The data in Table 10 reveals that among 100 respondents, the average score of 4.41, aligned with "strongly agree," indicates a significant level of agreement and satisfaction with the Barangay Appointment System's security features. This suggests that the respondents generally trust and feel secure with the system, demonstrating high satisfaction.

Table 11. Respondents' Assessment of the System in terms of Usability

USABILITY	Standard Deviation	Mean	Verbal Interpretation
1. The BAPS' features and functionalities suffice your needs to complete your task.	0.594588	4.5	Strongly Agree
2. The FAQs and step-by-step tutorials built into the system are helpful and usable.	0.698917	4.42	Strongly Agree
3. The BAPS system's user interface is intuitive and user-friendly.	0.635642	4.4	Strongly Agree
4. I am satisfied with the clarity and organization of the information presented in the BAPS system.	0.652811	4.41	Strongly Agree



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5. The information on the homepage's FAQs, modals, and barangay documents is presented clearly and concisely.	0.700937	4.44	Strongly Agree
General Weighted Mean:		4.43	Strongly Agree

In Table 11, the data reveals insights into the perceptions of 100 respondents regarding the Barangay Appointment System (BAPS). The mean score obtained is 4.43, which falls within the range associated with "strongly agree." This indicates that a significant majority of the respondents expressed a high level of agreement and satisfaction in terms of the system's usability and its intuitive interface.

Table 12. Respondents Assessment of the System in terms of Portability

PORTABILITY	Standard Deviation	Mean	Verbal Interpretation
1. The BAPS can be seamlessly accessed and used across different devices (e.g., desktop, laptop, mobile).	0.894653	4.26	Strongly Agree
2. The BAPS offers consistent functionality and user experience regardless of the device or platform used.	0.723069	4.32	Strongly Agree
3. The BAPS offers a responsive design that adapts well to various screen sizes and devices.	0.956583	4.21	Strongly Agree
4. The BAPS installation and setup process on different web servers is well-documented and user-friendly.	0.697687	4.41	Strongly Agree
5. I did not experience device crashes or instability while using BAPS.	0.714284	4.43	Strongly Agree
General Weighted Mean:		4.32	Strongly Agree

The table indicates that out of 100 respondents, the average score obtained was 4.32, corresponding to "strongly agree." This suggests that the majority of respondents



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expressed high agreement and satisfaction with the system's responsiveness, indicating positive perceptions and high satisfaction with its portability.

Table 13. Respondents Assessment of the System in terms of Reliability

RELIABILITY	Standard Deviation	Mean	Verbal Interpretation
1. The system fulfills the requirement for consistent reliability during normal operations.	0.749141	4.38	Strongly Agree
2. The system functions efficiently and remains readily available for use as needed.	0.627324	4.48	Strongly Agree
3. The system continues to function as intended despite the existence of hardware or software faults, the	0.827495	4.39	Strongly Agree
4. The system possesses the capability to restore the data that has been directly impacted and reinstate the intended state.	0.663325	4.38	Strongly Agree
5. The system maintains a robust incident response plan, including defined roles, responsibilities, and escalation procedures, to facilitate efficient and effective handling of reliability incidents.	0.670519	4.43	Strongly Agree
General Weighted Mean:		4.41	Strongly Agree

Based on the data presented in Table 10, the average score of 4.41 among 100 respondents, corresponding to "strongly agree," indicates a substantial level of agreement and satisfaction with the operational aspects and responsibilities of the Barangay Appointment System. These findings suggest that the respondents generally have trust and confidence in the system, leading to a high level of satisfaction.



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Table 14. Summary of All Gathered Data

QUESTION	
Functionality	Mean
The system's user interface is easy to understand and use.	4.59
The BAPS's features and functionalities are all present and working.	4.38
The system provided all of the information required to correctly complete the forms.	4.47
The system efficiently managed the clients queuing to reduce waiting time.	4.44
The system efficiently controlled the client's queue to reduce processing and waiting time.	4.57
The system successfully addresses all the designated tasks and fulfills the objectives set by the users	4.31
Efficiency	
The BAPS handled your appointment scheduling efficiently and effectively.	4.53
The system provided a convenient appointment schedule for requesting and claiming documents.	4.48
The BAPS can complete the required task at a sufficient level without any technological issues.	4.38
Security	
The security functionality of the BAPS system has a trustworthy authentication technique for user data protection.	4.44
The BAPS system has a function feature to ensure the user's information credibility.	4.37
The BAPS has security measures for fraudulent activities such as forgery and identity theft.	4.41
The BAPS implements strong access controls and authentication mechanisms to prevent unauthorized access.	4.46
The BAPS system clearly expresses its terms and agreement on using the website.	4.37
Usability	



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The BAPS' features and functionalities suffice your needs to complete your task.	4.5
The FAQs and step-by-step tutorials built into the system are helpful and usable.	4.42
The BAPS system's user interface is intuitive and user-friendly.	4.4
I am satisfied with the clarity and organization of the information presented in the BAPS system.	4.41
The information on the homepage's FAQs, modals, and barangay documents is presented clearly and concisely.	4.44
Portability	
The BAPS can be seamlessly accessed and used across different devices (e.g., desktop, laptop, mobile).	4.26
The BAPS offers consistent functionality and user experience regardless of the device or platform used.	4.32
The BAPS offers a responsive design that adapts well to various screen sizes and devices.	4.21
The BAPS installation and setup process on different web servers is well-documented and user-friendly.	4.41
I did not experience device crashes or instability while using BAPS.	4.43
Reliability	
The system fulfills the requirement for consistent reliability during normal operations.	4.38
The system functions efficiently and remains readily available for use as needed.	4.48
The system continues to function as intended despite the existence of hardware or software faults, the	4.39
The system possesses the capability to restore the data that has been directly impacted and reinstate the intended state.	4.38
The system maintains a robust incident response plan, including defined roles, responsibilities, and escalation procedures, to facilitate efficient and effective handling of reliability incidents.	4.43
Overall Average Mean	4.41

The survey indicates an overall average mean score of 4.41, which corresponds to a "strongly agree" rating in terms of verbal interpretation. In general, respondents expressed satisfaction with the system's functionality, efficiency, usability, portability, reliability, and security. However, despite the positive feedback received, there is still scope for further improvement in the system.



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Chapter V

SUMMARY OF FINDINGS, CONCLUSIONS, AND RECOMMENDATIONS

Summary

The determination to establish a Barangay Appointments System was motivated by the objective of optimizing and modernizing the appointment procedures within the barangay. The existing manual system proved to be laborious, ineffective, and deficient in terms of processing, transparency and accessibility. In light of these inadequacies, the proposed digital system endeavors to heighten efficiency through the provision of user-friendly features such as simplified appointment scheduling, reminders, and appointment tracking. This enhancement is designed to yield advantages for both residents and barangay officials, thereby making significant contributions to the overall efficiency and responsiveness of governance.

Residing within the barangay, individuals frequently encounter obstacles when endeavoring to procure essential documents. Predominant issues include insufficient awareness pertaining to requisite prerequisites and procedures, inefficient operational processes, protracted waiting periods, and an excessive burden of paperwork. These challenges are further compounded by the limited availability of barangay offices.

In response to the challenges, a meticulous selection process was undertaken, targeting residents of Barangay 177 based on comprehensive interviews conducted across three barangays. Analysis of the data uncovered a significant and consistent trend: Barangay 177 recorded the highest daily number of requesters, indicating a substantial volume of document requests. This finding implies that the current process of obtaining documents in Barangay 177 may pose inconveniences to its residents. By



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focusing on Barangay 177 residents as respondents, the study aims to ensure a more representative and robust sample, thereby enhancing the effectiveness of the research efforts.

Frequency and percentage distribution are valuable tools for summarizing statistical data, providing insights into the relative frequency of data points or groups. The mean, a measure of central tendency, is calculated by summing all values in a dataset and dividing by the total number of values, allowing researchers to interpret the average level of agreement on a Likert scale and assess the efficiency of the barangay appointment system's e-services. Range, on the other hand, measures variability and describes the spread of a dataset by subtracting the lowest value from the highest value. It is useful for comparing data sets and evaluating the level of agreement within the strongly agree or strongly disagree categories on the Likert scale. These statistical tools collectively contribute to a comprehensive understanding of the data and provide valuable insights for the study.

The proposed solution is the creation of the Barangay Appointment System, which aims to improve the process of obtaining barangay documents such as clearances and barangay IDs. This system will enhance organization and data analysis by providing a centralized platform for managing appointments and streamlining document processing. It will enable efficient tracking of document requests, facilitating informed decision-making and governance. Additionally, the system will promote transparency and accountability by providing residents with user-friendly access to appointment scheduling and request tracking. Overall, the implementation of the Barangay Appointment System will enhance efficiency, convenience, data management, and citizen-centric governance in Barangay 177, Camarin, Caloocan City.



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Findings

The survey conducted among 100 participants revealed a gender disparity within the sample, with 64% of respondents being male and 36% being female. The majority of document requesters belonged to the 18-24 age group, accounting for 72% of respondents. Students represented the largest employment group at 58%, followed by employees at 33%, which included 5 % of barangay officials, and job seekers at 9%. The most commonly requested documents were Barangay Certificate/Clearance (82%), followed by Barangay Indigency (53%), Barangay Identification (50%), and Business Clearance (26%).

The Barangay Appointment System efficiently manages the queuing process for document requests through various features. Clients can schedule appointments online, reducing queues, and receive automated reminders. The system incorporates a virtual queue system, allowing clients to monitor their position without physically waiting. It also optimizes resource allocation and streamlines the document processing workflow, minimizing delays. Data analysis helps predict peak hours and allocate resources effectively, enhancing efficiency, reducing wait times, and improving client satisfaction.

The system is designed to be accessible and user-friendly on both mobile devices and personal computers. It uses responsive web design to adapt its layout and functionality to different screen sizes. The mobile interface caters to touch-based interactions, while the desktop interface provides a more expansive view. Users can access the system seamlessly on their preferred device, accommodating residents and barangay officials on the go or working from their computers.

The survey findings indicate high levels of satisfaction among the respondents regarding the system's functionality, efficiency, usability, portability, reliability, and security. The system received positive ratings in all aspects, with an average mean



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score of 4.41. However, portability had the lowest average score. Despite the positive feedback, there is a continuous focus on enhancing the system to meet evolving user needs and expectations. Recognizing opportunities for growth and refinement ensures the system's ongoing performance elevation.

Conclusions

1. The researchers conclude that there are several significant patterns among the participants. Firstly, there exists a noticeable gender imbalance, with males comprising the majority at 64% while females represent the minority at 36%. Regarding age groups, it is evident that the 18-24 age bracket dominates document requests in the barangay, accounting for 72% of the overall sample.

Furthermore, upon analyzing the employment status of the participants, the majority identified themselves as students, constituting 58% of the total sample. This implies that students are the primary individuals seeking document requests in Barangay 177, Camarin. Additionally, the study highlights the most commonly requested documents, with the Barangay Certificate/Clearance being in highest demand, indicated by 82% of respondents requiring it.

2. The researchers conclude that the implementation of the Barangay Appointment System will effectively handles the document request queuing process by utilizing a range of features. Through online appointment scheduling, clients can efficiently manage their visits, resulting in reduced queues, while automated reminders help them stay on track. The system integrates a virtual queue system, enabling clients to monitor their position without the need for physical waiting. Moreover, it optimizes resource allocation and streamlines the workflow for document processing, resulting in minimized delays. By leveraging data analysis, the system can anticipate peak hours and allocate resources,



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accordingly, thereby enhancing overall efficiency, reducing wait times, and ultimately improving client satisfaction.

3. The researchers conclude that the system has been thoughtfully designed to prioritize accessibility and user-friendliness on mobile devices and personal computers. The system effortlessly adapts its layout and functionality by utilizing responsive web design to accommodate various screen sizes. The mobile interface facilitates intuitive touch-based interactions, while the desktop interface provides a more comprehensive view recommended for system administrators. This seamless accessibility ensures that residents and barangay officials can effortlessly access the system on their preferred devices, whether on the move or working from their computers. Overall, the system's inclusive design guarantees a user-friendly experience for all users, promoting convenience and efficiency in accessing its features.

4. The researchers conclude that the respondents expressed high levels of satisfaction across various aspects of the system, including functionality, efficiency, usability, portability, reliability, and security. The functionality section received an average score of 4.46, indicating that the system's performance aligned with the respondents' expectations. Similarly, the efficiency section achieved an average score of 4.46, demonstrating the system's effective and efficient operation. The security section obtained a general weighted mean of 4.41, highlighting the respondents' trust in the system's security measures. The usability section scored an average of 4.43, indicating that the system's features met the respondents' expectations. Regarding portability, the system achieved an average score of 4.32, suggesting minimal issues with device crashes while using the system. Lastly, the reliability section received an average score of 4.41, signifying the respondents' trust in the system for storing data. Among all the aspects evaluated, portability obtained the lowest average score, emphasizing the need



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to improve responsiveness, irrespective of the "strongly agree" consensus. Although the survey yielded positive feedback, it is crucial to recognize the continuous pursuit of improvement.

Recommendations

1. Implementation of mobile application: Development of a mobile application for the barangay appointment system will help on creating more accessible options and easy access for the users of the system.
2. Launching of online payment services: Implementation of online payment such as e-wallet and bank transfer will improve the convenience and efficiency of the processing of documents.
3. Further enhancement of e-services: Additional document services for the appointment system will make the system create more user engagement and improve the overall quality of efficiency of the system.
4. Handling of bigger government institutions: By improving and developing the appointment system, providing e-services for much larger municipalities or government institutions will be the next goal of the system.
5. Security enhancement: Investment in developing better security for the user's data, protection in identity thefts, scams, fraud, data breach, and hacking attempts should be a priority to protect the confidentiality of the user's personal data.
6. Improvement user interface: Consistent improvement of the user interface, graphical design and navigation will aid on increase user engagement in the system and will also improve the quality performance of the system.



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7. Regular System Updates: Constant system updates and maintenance will prevent server crashes, network errors, and optimizing the BAPS with the current technology will help the system stay relevant and be a useful tool for users for a long time.

By following these recommendations, BAPS will be able to improve the functionality, reliability, efficiency, usability, security, and use of QR Code technology, thereby achieved a high satisfactory rating from the users that will result to a better satisfaction level in the service within the government institutions.



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APPENDICES

Appendix 1: Survey Questionnaire

Polytechnic University of the Philippines

Institute of Technology

Greetings!

We are third-year students from the Polytechnic University of the Philippines - Sta. Mesa Manila, currently pursuing a Diploma in Information Communication Technology. As part of our academic curriculum, we are currently engaged in a capstone project entitled: **BAPS: BARANGAY APPOINTMENT SYSTEM FOR LOCAL BARANGAYS**. The primary objective of this system is to offer residents a more efficient and convenient method for requesting documents at the barangay.

We kindly request a moment of your valuable time to participate in our survey regarding the system. Your feedback will play a vital role in refining the Barangay Appointment System and ensuring a high-quality user experience.

In adherence to Republic Act 10173 or the Data Privacy Act of 2012, all information shared in the survey will be treated with utmost confidentiality and solely used for academic purposes.

We sincerely appreciate your time and consideration!

Researchers:

Bayaban, Christian Jay Padron, Neil Patrick M. Magnaye, Jeremy I.

Escanlar, Mark V. Priel, Ralph Allendrey O.

Part I. Profile of the Respondents

Instruction: Please place a check on the space that corresponds to your answer.

1. Gender:

_____ Male



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_____ Female

2. Age:

_____ 18-24

_____ 25-34

_____ 35-44

_____ 45 and above

3. Type of client:

_____ Student

_____ Job seeker

_____ Employee

4. Type of document requested:

_____ Barangay Clearance/Certificate

_____ Barangay Certificate of Indigency

_____ Barangay Identification

_____ Barangay Business Clearance

Part II. Software Quality

Directions: Using the scale given below, place a check mark in the column to rate the quality of the software.

5 – **Strongly Agree**, 4 – **Agree**, 3 – **Neutral/Uncertain**, 2 – **Disagree**, 1 – **Strongly Disagree**

FUNCTIONALITY	5	4	3	2	1
The system's user interface is easy to understand and use.					
The BAPS's features and functionalities are all present and working.					
The system provided all of the information required to correctly complete the forms.					
The system efficiently managed the clients queuing to reduce waiting time.					
The system efficiently controlled the client's queue to reduce processing and waiting time.					
The system successfully addresses all the designated tasks and fulfills the objectives set by					



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the users					
	5	4	3	2	1
EFFICIENCY					
The BAPS handled your appointment scheduling efficiently and effectively.					
The system provided a convenient appointment schedule for requesting and claiming documents.					
The BAPS can complete the required task at a sufficient level without any technological issues.					
	5	4	3	2	1
SECURITY					
The security functionality of the BAPS system has a trustworthy authentication technique for user data protection.					
The BAPS system has a function feature to ensure the user's information credibility.					
The BAPS has security measures for fraudulent activities such as forgery and identity theft.					
The BAPS implements strong access controls and authentication mechanisms to prevent unauthorized access.					
The BAPS system clearly expresses its terms and agreement on using the website.					
	5	4	3	2	1
USABILITY					
The BAPS' features and functionalities suffice your needs to complete your task.					
The FAQs and step-by-step tutorials built into the system are helpful and usable.					
The BAPS system's user interface is intuitive and user-friendly.					



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I am satisfied with the clarity and organization of the information presented in the BAPS system.					
The information on the homepage's FAQs, modals, and barangay documents is presented clearly and concisely.					
	5	4	3	2	1
PORTABILITY					
The BAPS can be seamlessly accessed and used across different devices (e.g., desktop, laptop, mobile).					
The BAPS offers consistent functionality and user experience regardless of the device or platform used.					
The BAPS offers a responsive design that adapts well to various screen sizes and devices.					
The BAPS installation and setup process on different web servers is well-documented and user-friendly.					
6. I did not experience device crashes or instability while using BAPS.					
	5	4	3	2	1
RELIABILITY					
The system fulfills the requirement for consistent reliability during normal operations.					
The system functions efficiently and remains readily available for use as needed.					
The system continues to function as intended despite the existence of hardware or software faults, the					
The system possesses the capability to restore the data that has been directly impacted and reinstate the intended state.					



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The system maintains a robust incident response plan, including defined roles, responsibilities, and escalation procedures, to facilitate efficient and effective handling of reliability incidents.

Appendix 2: Interview Questionnaire

BAPS: BARANGAY APPOINTMENT SYSTEM

FOR LOCAL BARANGAYS

A Capstone Project

Presented to the Faculty of the Institute of Technology

Polytechnic University of the Philippines

Sta. Mesa, Manila

DIRECTION: Please write your answers and their corresponding question numbers at the next page of this paper. Also, please sign this paper with your name and signature on it after answering all the questions below.

(You can answer the questions in a brief and concise response. Thank you!)

Questions:

1. How often do the residents ask for certain documents?
2. Who are the most frequent people asking for certain documents? (Students, Businesses, Employees, etc.)
3. What is the average/estimate of individuals who ask or inquire in the barangay?
4. How many kinds of documents that [barangay] are offering to their residents? Could you list the documents?
5. In question 4. What are the purposes or uses of each of these certain documents are for? Could you list it down?
6. Are there specific requirements needed to request certain documents?



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7. How long does it take to apply or request a document?
8. How long does it take to finish and claim the documents they requested?
9. How do they inform the requestor if their documents are ready for pickup?
10. Is there a fee for requesting certain documents? If yes, how much is it, and how will they pay for it?

Appendix 3: Plagiarism Scan Result (Turnitin)

BAPS: Barangay Appointment System for Local Barangays

ORIGINALITY REPORT

8%

SIMILARITY INDEX

7%

INTERNET SOURCES

1%

PUBLICATIONS

5%

STUDENT PAPERS



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Appendix 4: Grammar and Plagiarism Scan Result (Grammarly)



Report: BAPS: BARANGAY APPOINTMENT SYSTEM FOR LOCAL BARANGAYS IN CALOOCAN CI...

BAPS: BARANGAY APPOINTMENT SYSTEM FOR LOCAL BARANGAYS IN CALOOCAN CITY

by Frinz Christopher Pile

General metrics

90,869
characters

13,224
words

1198
sentences

52 min 53 sec
reading
time

1 hr 41 min
speaking
time

Score

99

This text scores better than 99%
of all texts checked by Grammarly

Writing Issues

169
Issues left

40
Critical

129
Advanced

Plagiarism

3
%

33
sources

3% of your text matches 33 sources on the web
or in archives of academic publications

Report was generated on Monday, Jul 17, 2023, 09:17 PM

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APPENDIX 5: Interview Documentation:



Nheli Celso-Brgy Kagawad of Barangay Manggahan, Pasig City



Jhenjhen Cadalin- Brgy Kagawad of Barangay 177 Camarin, Caloocan City



Ludivino P. Reyes- Brgy Kagawad of Brgy. 262 Zone 24 Tondo, Manila



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Gather experience in this potential work and further enhance the skills that I have acquired through my education. I want to be able to exert my best at a given task in order to improve myself and become a better version of me every day.

WORK EXPERIENCE

DECEMBER 2019 – FEBRUARY 2019

WORK IMMERSION TRAINEE, ACLC MAIN OFFICE

- 59 Panay Ave, Diliman, Quezon City, Metro Manila
- (300 hours Training)

MAY 2023 – AUGUST 2023

IT INTERN, HYT FOUNDATION INC.

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- (600 hours Training)

EDUCATION

OCTOBER 2020 - PRESENT

DIPLOMA IN INFORMATION COMMUNICATION TECHNOLOGY,
POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

2018-2020

SENIOR HIGH SCHOOL (TECH-VOC ICT), ACLC COLLEGE OF MANILA

- WITH HIGH HONORS

2014-2018

JUNIOR HIGH SCHOOL, RAMON MAGSAYSAY HIGH SCHOOL MANILA

SKILLS

COMPUTER HARDWARE/SOFTWARE

- Proficiency on Microsoft Office
- Basic Fundamentals on HTML, CSS, JAVA, and SQL
- Basic Hardware Configuration
- Software Installation
- Typing Speed of 40 WPM



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To apply all my knowledge, to exploit and share my skillfulness, and talents to others as well as to be exposed and be productive in the factual setting of the real competitive work.

WORK EXPERIENCE

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WORK IMMERSION TRAINEE, OUR LADY OF FATIMA, LAGRO BRANCH

- 1118, 1 Esperanza, Novaliches, Lungsod Quezon, 1118 Kalakhang Maynila
- (80 hours Training)

DECEMBER 2022 – JANUARY 2023

TESDA TRAINEE, MINDTECH TRAINING AND DEVELOPMENT INSTITUTE INC.

- #163 Dr. Sixto Antonio Avenue Rosario, Pasig, Philippines
- (56 days Training)

MAY 2023 – AUGUST 2023

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EDUCATION

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SENIOR HIGH SCHOOL (STEM), LAGRO HIGH SCHOOL

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2014-2018

JUNIOR HIGH SCHOOL, CIELITO ZAMORA JUNIOR HIGH SCHOOL

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SKILLS

COMPUTER HARDWARE/SOFTWARE

- Computer Assembly/Disassembly
- Basic Hardware Configuration
- Proficiency on Microsoft Office
- Visual Studio
- Basic skills on C++
- Soft skills on Java
- Software Installation
- Network Configuration



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- Basic fundamentals on HTML, CSS and Bootstrap

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Secure a responsible career opportunity to fully utilize my training and skills, while making a significant contribution to the success of the company.

WORK EXPERIENCE

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WORK IMMERSION TRAINEE, ADAMSON UNIVERSITY

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2014-2018

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SKILLS

COMPUTER HARDWARE/SOFTWARE

- Proficiency on Microsoft Office
- Basic Fundamentals on Database, HTML, CSS, and JavaScript
- Computer Assembly/Disassembly
- Basic Hardware Configuration
- Software Installation
- Typing Speed of 58 WPM



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To apply all my knowledge, to exploit and share my skillfulness, and talents to others as well as to be exposed and be productive in the factual setting of the real competitive work.

WORK EXPERIENCE

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JUNIOR HIGH SCHOOL, LAGRO HIGH SCHOOL

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SKILLS

COMPUTER HARDWARE/SOFTWARE

- Proficiency on Microsoft Office
- Basic Fundamentals on HTML, CSS, and JavaScript
- Computer Assembly/Disassembly
- Basic Hardware Configuration
- Software Installation
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To utilize my knowledge, skills, and talents to benefit others and thrive in a challenging professional environment, where I can contribute and excel.

WORK EXPERIENCE

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SENIOR HIGH SCHOOL (ABM), LIGAO NATIONAL HIGH SCHOOL

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2014-2018

JUNIOR HIGH SCHOOL, LIGAO NATIONAL HIGH SCHOOL

- WITH HONORS

SKILLS

COMPUTER HARDWARE/SOFTWARE

- Proficient with Microsoft Office Applications
- Computer Literate
- Able to communicate through written and oral forms
- Has a positive attitude and excels at collaborative work
- Fundamentals on HTML, CSS, and JavaScript
- Fundamentals on HTML, CSS and Bootstrap
- Basic knowledge in Adobe Photoshop and Canva
- Software Installation
- Basic Hardware Configuration